

# Overcoming the 3 dB squeezing extraction limit in silicon carbide microcombs with a photonic molecule: supplemental document

This supplemental document provides supporting content for the primary manuscript: the material dispersion models and their sources, the finite-element mode-solver settings with a mesh-convergence study, the dispersion-extraction procedure, the normalization and numerics of the modal Lugiato–Lefever solver including the co-moving-frame treatment of the drifting crystal, the derivation and analytic validation of the linearized multimode quantum model, and numerical convergence studies for all quoted quantum quantities.

## 1. MATERIAL DISPERSION MODELS

Both material models come from the refractiveindex.info database, which is released to the public domain under CC0 [1]. The 4H-SiC ordinary-ray index uses the Sellmeier model of Wang *et al.* [2],

$$n_o^2(\lambda) = 1 + \frac{0.20075 \lambda^2}{\lambda^2 + 12.07224} + \frac{5.54861 \lambda^2}{\lambda^2 - 0.02641} + \frac{35.65066 \lambda^2}{\lambda^2 - 1268.24708}, \quad (\text{S1})$$

with  $\lambda$  in micrometres, valid from 0.4047 to 5  $\mu\text{m}$ . The chip normal is the crystal  $c$  axis, so the fundamental TE mode is polarized in the ordinary plane and Eq. (S1) applies. The silica cladding uses the Malitson model [3],

$$n^2(\lambda) = 1 + \frac{0.6961663 \lambda^2}{\lambda^2 - 0.0684043^2} + \frac{0.4079426 \lambda^2}{\lambda^2 - 0.1162414^2} + \frac{0.8974794 \lambda^2}{\lambda^2 - 9.896161^2}. \quad (\text{S2})$$

At 1.55  $\mu\text{m}$  these give  $n_o(\text{SiC}) = 2.5644$  and  $n(\text{SiO}_2) = 1.4440$ . The Kerr index  $n_2 = 6.9 \times 10^{-19} \text{ m}^2/\text{W}$  is taken from measurements on 4H-SiC [2, 4].

## 2. FEM MODE SOLVING AND CONVERGENCE

Modes are computed with the open-source solver `femwell` (v0.x, scikit-fem backend [5]) on triangular meshes generated by Gmsh [6]. The computational window spans  $6.4 \times 4.9 \mu\text{m}^2$  around the core, with a core-region resolution of 35 nm for the geometry sweep and 40 nm for the selected-geometry runs, and graded coarsening into the cladding. Two modes are requested per solve and the TE-polarized mode is selected by its TE fraction, which exceeds 0.99 for all geometries.

Table S1 reports the mesh study at  $\lambda = 1.55 \mu\text{m}$  for the selected geometry. With second-order elements the effective index is converged to below  $10^{-7}$  between the two finest meshes, and the production setting (order 2, 40 nm) is indistinguishable from the finest run at that precision. The geometry sweep of the main text uses first-order elements for speed; the resulting absolute offset of about  $6 \times 10^{-4}$  in  $n_{\text{eff}}$  varies smoothly with wavelength and therefore has a negligible effect on the dispersion coefficients, which depend only on differences across the resonance grid.

## 3. DISPERSION EXTRACTION

The effective index is sampled at 17 wavelengths from 1.30 to 1.85  $\mu\text{m}$  and fitted with a degree-six polynomial in a normalized frequency variable; the maximum fit residual is below  $5 \times 10^{-8}$ . Resonances of a ring of length  $L = 2\pi R$  solve  $\beta(\omega_m)L = 2\pi m$  with  $\beta = n_{\text{eff}}\omega/c$ , found by bracketed root finding on the fitted model for each azimuthal order. The radius is chosen so that the FSR at 1.55  $\mu\text{m}$  equals 350 GHz, giving  $R = 50.06 \mu\text{m}$ . A quartic fit over  $|\mu| \leq 40$  yields  $D_2/2\pi = 6.89 \text{ MHz}$ ,  $D_3/2\pi = -63.4 \text{ kHz}$ , and  $D_4/2\pi = -148 \text{ Hz}$ . The LLE and quantum stages use the Taylor form of  $D_{\text{int}}$ , which agrees with the directly solved resonance grid to within the fit residual over the simulated band. Bending corrections at  $R \approx 50 \mu\text{m}$  are neglected; for this high-confinement core the bend-induced index shift is far smaller than the geometric dispersion tuning range explored in the design map.

**Table S1.** Mesh convergence of  $n_{\text{eff}}$  at 1.55  $\mu\text{m}$  ( $w = 1.85 \mu\text{m}$ ,  $h = 500 \text{ nm}$ , oxide clad).

| core resolution (nm) | element order | elements | $n_{\text{eff}}$ |
|----------------------|---------------|----------|------------------|
| 60                   | 1             | 3 426    | 2.3068171        |
| 50                   | 1             | 3 426    | 2.3067888        |
| 40                   | 1             | 4 612    | 2.3072447        |
| 30                   | 1             | 6 326    | 2.3075482        |
| 50                   | 2             | 3 426    | 2.3081254        |
| 40                   | 2             | 4 612    | 2.3081259        |
| 30                   | 2             | 6 326    | 2.3081259        |

#### 4. LUGIATO–LEFEVER SOLVER

##### A. Normalization

Time is measured in units of  $2/\kappa$ , detunings as  $\zeta_\mu = 2(\delta_0 + D_{\text{int}}(\mu))/\kappa$ , and the field is scaled so that the self-phase-modulation rate equals  $(\kappa/2)|\psi|^2$ . In this convention the pump term is  $f^2 = 8\eta g_0 P_{\text{in}}/(\kappa^2 \hbar \omega_0)$  with  $\eta = 1/2$  at critical coupling, and the intracavity photon number of mode  $\mu$  is  $\kappa|\phi_\mu|^2/(2g_0)$  with  $\phi_\mu$  the modal amplitude in the azimuthal normalization. The single-soliton existence limit at fixed pump is  $\zeta_0^{\text{max}} = \pi^2 f^2/8$ , which for  $f^2 = 9$  gives 11.1, in agreement with the observed annihilation point.

##### B. Numerics

Equation (1) of the main text is integrated with a symmetrized (Strang) split-step scheme: the linear part, including the pump, is advanced exactly in the modal basis, and the Kerr part is an exact phase rotation in the azimuthal basis. Production runs use  $N = 192$  modes and  $\Delta t = 0.002$ . Table S2 shows that halving the step or increasing the grid changes the comb-line amplitudes by parts in  $10^4$ .

**Table S2.** LLE numerical convergence at  $\zeta_0 = 6.5$  (modal amplitude of comb line  $\mu = 2$ , peak field, drift velocity).

| $N$ | $\Delta t$ | $ \phi_2 $ | $\max  \psi $ | $v$                       |
|-----|------------|------------|---------------|---------------------------|
| 192 | 0.004      | 0.342006   | 3.861400      | $-2.21707 \times 10^{-2}$ |
| 192 | 0.002      | 0.341966   | 3.861302      | $-2.21732 \times 10^{-2}$ |
| 192 | 0.001      | 0.341956   | 3.861196      | $-2.21738 \times 10^{-2}$ |
| 256 | 0.002      | 0.341966   | 3.841014      | $-2.21732 \times 10^{-2}$ |

##### C. Co-moving frame

With odd-order dispersion the crystal drifts at constant velocity  $v$ , so modal phases rotate as  $\psi_\mu(t) = \psi_\mu(0)e^{-i\mu v t}$ . We converge each state in a frame where a linear detuning term  $-v\mu$  absorbs the drift, iterating between (i) relaxation under the modified detuning and (ii) measurement of the residual drift from the modal phase slopes. Three to four iterations reduce the stationarity residual  $\max_\mu |\dot{\psi}_\mu|$  from order one to below  $10^{-3}$ . The converged  $v$  equals the comb repetition-rate shift,  $\delta f_{\text{rep}} = v(\kappa/2)/2\pi$ , and vanishes to machine precision when  $D_3$  is set to zero, an internal consistency check of the solver. States whose co-moving residual exceeds  $10^{-2}$  are classified as breathing and excluded from the quantum analysis. The odd/even modal power contrast of converged crystals is about 220 dB, confirming exact  $C_2$  symmetry.

## 5. LINEARIZED QUANTUM MODEL

### A. Drift matrix

Substituting  $\hat{a}_\mu = A_\mu + \hat{b}_\mu$  into the intracavity Kerr Hamiltonian and keeping quadratic terms yields Eq. (2) of the main text. In the doubled basis  $\hat{\mathbf{v}} = (\hat{b}_{-M}, \dots, \hat{b}_M, \hat{b}_{-M}^\dagger, \dots, \hat{b}_M^\dagger)^\top$  the dynamics is  $\dot{\hat{\mathbf{v}}} = \mathbf{M}\hat{\mathbf{v}} + \Xi\hat{\mathbf{v}}_{\text{in}}$  with

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{A}^* \end{pmatrix}, \quad \mathbf{A}_{qk} = \delta_{qk} \left( -\frac{\kappa_q}{2} - i\Delta_q \right) + 2ig_0 R_{qk}, \quad \mathbf{B}_{qk} = ig_0 P_{qk}, \quad (\text{S3})$$

where  $\Delta_q$  is the detuning of fluctuation mode  $q$  from the comb grid in the co-moving frame,  $\Delta_q = (\kappa/2)(\zeta_q - vq)$  in normalized units.  $R$  is Hermitian and  $P$  symmetric by construction. For a 2-FSR crystal,  $P_{qk}$  and  $R_{qk}$  vanish unless  $q$  and  $k$  have equal parity, so the odd fluctuation sector decouples from the even one and carries no classical field.

### B. Detection model

Each mode couples to an unmonitored intrinsic bath at rate  $\kappa_q^i$  and a monitored collection port at rate  $\kappa_q^c$ . Solving in the frequency domain gives the port scattering matrix, and the symmetrized quadrature spectral matrix of the detected modes follows from vacuum inputs on all ports, averaged over  $\pm\omega$  sidebands as measured by homodyne detection. Squeezing values are eigenvalues of this matrix, normalized so vacuum equals unity; supermodes are its eigenvectors; and two-mode logarithmic negativities use the standard covariance formula [7]. For the photonic molecule, odd modes take  $\kappa^i = \kappa$  (all bare loss unmonitored) and  $\kappa^c = \kappa_P$ , after adiabatic elimination of a strongly damped auxiliary ring ( $\kappa_{\text{aux}} \gg J$ ) whose Purcell rate is  $\kappa_P = 4J^2/\kappa_{\text{aux}}$ .

### C. Analytic validation

The implementation passes three checks. (i) With no comb the output spectral matrix equals the identity at all frequencies to machine precision ( $< 3 \times 10^{-16}$ ). (ii) For a cw-pumped resonator the drift eigenvalues of the  $(\mu, -\mu)$  pair reproduce the analytic Bogoliubov result  $\lambda = -\kappa/2 \pm \sqrt{g^2 - \bar{\delta}^2}$  with  $g$  the parametric gain and  $\bar{\delta}$  the shifted detuning, to  $10^{-14}$ . (iii) At critical coupling the computed output squeezing never exceeds the analytic 3 dB bound, the Heisenberg product  $\min V \times \max V \geq 1$  holds on all frequency grids, and over-coupling raises the detected squeezing as expected.

### D. Truncation study

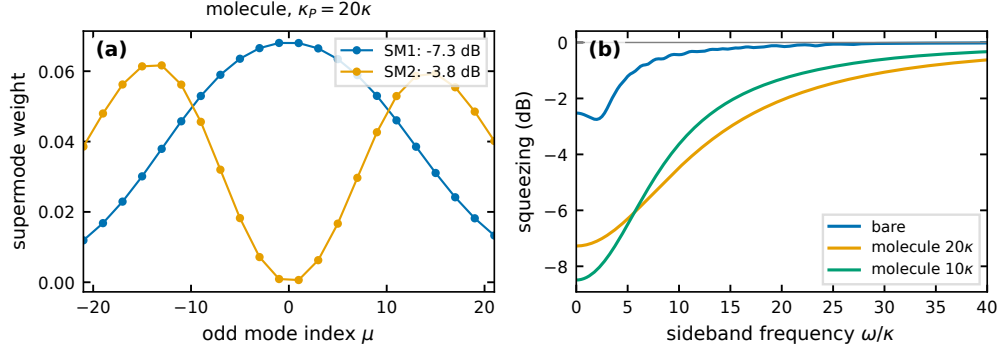
Production runs keep fluctuation modes  $|\mu| \leq 30$  and classical comb modes  $|\mu| \leq 60$ . Table S3 shows the dependence of the peak squeezing on the truncation  $M$ . The bare-device value oscillates by about 0.1 dB, limited by the edge of the truncated lattice; the molecule value converges from below with a change of 0.22 dB from  $M = 30$  to  $M = 40$ . We therefore quote molecule squeezing with a numerical uncertainty of about 0.2 dB; the production values are slightly conservative.

**Table S3.** Peak squeezing versus fluctuation-mode truncation at  $\zeta_0 = 6.5$ .

| $M$ | bare (dB) | molecule $\kappa_P = 20\kappa$ (dB) |
|-----|-----------|-------------------------------------|
| 20  | -2.886    | -6.616                              |
| 30  | -2.751    | -7.275                              |
| 40  | -2.963    | -7.498                              |

## 6. SUPPLEMENTARY FIGURE

Figure S1 shows the supermode structure at the molecule drop port and a direct comparison of bare and extracted squeezing spectra. The two-supermode structure of the bare device survives Purcell extraction, and the spectra broaden as the odd-mode linewidths grow.



**Fig. S1.** (a) Weights of the two dominant squeezed supermodes at the drop port for  $\kappa_P = 20\kappa$ . (b) Optimal squeezing spectra of the bare device and the molecule at  $\kappa_P = 10\kappa$  and  $20\kappa$  ( $\zeta_0 = 6.5$ , pump 8.3 mW).

## 7. REPRODUCING THE RESULTS

The simulation code is openly available on GitHub at [github.com/Tanvir-Mahmud-Mahim/sic-molecule-squeezer](https://github.com/Tanvir-Mahmud-Mahim/sic-molecule-squeezer) (Apache-2.0 license), and the complete raw data together with the validation testbench and its reference outputs are archived on Zenodo at [doi:10.5281/zenodo.21471674](https://doi.org/10.5281/zenodo.21471674) (CC-BY-4.0). The released package contains the complete pipeline: `materials.py` (Sellmeier models), `fem_modes.py` and `sweep_fem.py` (FEM), `dispersion.py` (resonance grid), `lle.py` and `exp_lle.py` (mean field), `quantum.py`, `exp_quantum.py`, and `exp_addendum.py` (quantum stage), `validate_quantum.py` and `convergence_checks.py` (verification), and `make_numbers.py`, which regenerates every number quoted in the manuscript directly from the simulation outputs. The testbench pass criteria are: vacuum output equal to the identity to better than  $10^{-15}$ ; Bogoliubov drift eigenvalues matching the closed form to  $10^{-13}$ ; critically coupled output squeezing never exceeding the 3.01 dB bound; Heisenberg product  $\min V \times \max V \geq 1$  on all frequency grids; and the convergence tolerances of Tables S2 and S3.

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